

MARGARITA MONTAÑEZ DAVENPORT

Clinical Systems & Health Informatics | Healthcare Integration | FHIR/HL7

margarita.m.davenport@gmail.com · mmdavenport.github.io · github.com/mmdavenport · LinkedIn

SUMMARY

Clinical systems specialist focused on healthcare integration, with thirty years in clinical practice and a health data platform built from the ground up. Designed and built a FHIR R4 integration pipeline on Open Integration Engine, mapped wearable device data to LOINC-coded Observations via the Google Health API, and built the consent and identity layer underneath both. BS in Management Information Systems & Business Analytics, Colorado State University. Health Informatics certificate, The Johns Hopkins University.

SKILLS

Product	Discovery and roadmapping · requirements and success metrics · cross-functional delivery · 0-to-1
Front end	React · TypeScript · design systems · Storybook · tokens and variants · accessibility (WCAG AA)
Decentralized	Solidity · Hardhat · ERC-20 and ERC-721 · on-chain consent and attestation · EIP-6963 wallet integration · Merkle proofs · IPFS
Data	SQL · Python · Pandas · R · Tableau · Jupyter · schema design · ETL
Health systems	HL7v2 · MLLP · interface engine channels · FHIR R4 · OMOP · EHR mapping · clinical workflow design
Clinical systems	Dentrix · Denticon · Eaglesoft · Open Dental · MacPractice · Archy · tab32 · SoftDent

EXPERIENCE

Founder & Product Lead · Future Systems Lab LLC 2023 – Present

Own product strategy and delivery for a health data platform. Built a wearable data integration against the Google Health API — OAuth authorization, FHIR R4 Observation mapping with LOINC coding, and scoped time-limited access tokens — behind a consent layer where the participant holds a signed, verifiable record of every grant and revocation. Wrote the gap analysis, architecture, and staged adoption roadmap. Eleven contracts authored, deployed, and source-verified on Ethereum Sepolia. Ran the security audit.

Live demonstration: demo.futuresystemslab.io

HL7v2→FHIR Integration: built an interface engine channel on Open Integration Engine (OIE fork of Mirth Connect 4.5.2) — HL7v2 ORU^R01 messages transformed to FHIR R4 Observations via LOINC-mapped JavaScript transformer; 6/6 synthetic test messages passing with AA acknowledgment over MLLP. evidence.futuresystemslab.io/hl7-fhir

Registered Dental Hygienist · Carolina Dental Arts 06/2025 – Present

Practicing clinically through the hygienist shortage; designed and implemented an assisted double-hygiene model, running two columns of daily production.

Practice Operations Lead & Clinical Lead · Riccobene Associates Family Dentistry 2019 – 2025

Traveled between practices in one of the Southeast's largest dental service organizations, standing up hygiene departments in offices that had no provider — building the workflow and compensation model at each site, then handing it off. Apex and Durham roughly a year each. Documented at Durham: per-provider production \$380,710 to \$624,640 annualized from ten months, a 64% increase across four documented years, with department share of office production rising 30.4% to 38.1% — real growth in total office production. Documented figures are from Durham; earlier sites predate a practice-management migration.

Founder & Clinical Director · SunCodes Holistic Health 2018 – 2023

Founded and ran a consumer health product line end to end, formulation through retail, then moved it direct-to-consumer in 2020.

OPEN SOURCE

[bcgov/aries-vcr PR #811 \(merged\)](#) · [openmrs/fhir2 PR #577](#) and [openmrs/fhir2-ig PR #63](#) (under review)

EDUCATION AND CERTIFICATIONS

B.S., Management Information Systems & Business Analytics · Colorado State University, 2026

Registered Dental Hygienist, North Carolina — license #7177, since 2003 · Certified Blockchain Healthcare Professional · Certified Smart Contract Auditor (Blockchain Council) · Google UX Design · Health Informatics, Johns Hopkins

Member: IEEE · INCOSE · ISC² · AMIA · HIMSS · PDMA